

ED Research System — Handoff (Freeze Snapshot)

Timestamp: 2025-08-24 12:42 UTC

Status: Phase 1 (Operational Infrastructure) — COMPLETE; Phase 2 (Clinical Integration)

Purpose

This handoff freezes the current working state for safe transfer. It captures architecture, invariants, what is implemented, how to run, paths, quickchecks, and known limits. On advice-only; no gates are flipped by ML.

Architecture & Invariants

Core Components

- Workflow layer: WorkflowState + TinyCritics (cold-start smoke passed).
- Tracker core: EquipmentRepository, MovesLogRepository, SOPRegistry, QRService, TrajectoryEngine
- UI: Streamlit app (split helpers + wrapper), with adjustable thresholds for equipment vitals overdue.
- ML bridge: advice-only MLP loader + scorer (bundle autodetect, thresholded, calibrated)
- Gate engine: wired and sanity-checked (TTL_MAX_OK, GATES_WIRED_OK, PHASE2_BUNDLE_SIZE_LIMIT)

Invariants (observed)

- CONFIG is frozen after bootstrap; only MODEL_BUNDLE_PATH may be set later.
- No top-level side effects beyond path creation.
- Paths preserve CSV/dir semantics: EQUIPMENT_STATUS_PATH, EQUIPMENT_MOVES_LOG_PATH, SOP_REGISTRY_PATH, QR_OUTPUT_DIR, EVENT_LOG_PATH.
- QR: generate + scan-to-update fallback via manual payload is present in UI; decoding optional.
- SOP auto-pull: offline-safe surface; returns summary if network/libs absent.
- Overdue/alerts: equipment and lingering vitals use adjustable thresholds in the UI

What's Implemented (Evidence)

Tracker / QR / SOPs

- Moves logging: TrackerService.log_move writes to moves log; EquipmentRepository.read always return a DataFrame.
- QR generate and manual fallback path updates moves (payload like 'id=pump-777').
- SOP auto-pull function refresh_sop_registry(CONFIG, base_url=...) present; offline verified.
- Finder & analytics: find_equipment, movement_stats (per-equipment counts, routes).

UI

- Split helpers: run_ui_header_controls (contains 'Vitals overdue (min)' + 'Overdue (min)'), panels for Equipment, QR, SOPs, Actions & Critic, Movement Analytics; wrapper run_ui them.

ML (advice-only)

- Bundle autodetect + load (ed_phase2_model*_patched(1).joblib).
- Feature shim over MIMIC-IV-ED demo; scoring produces advice_gatepos_mlp.csv (222 rows run).
- Calibrator-safe fallback; threshold respected; no gate flips.

Gate Engine Sanity

- TTL_MAX_OK: OK; GATES_WIRED_OK: OK; PHASE2_BUNDLE_SMOKE_OK: OK.

How to Run (Safe Defaults)

Flags

- CONFIG is frozen; do not assign RUN_* in-place post-freeze. Use an overlay for test

```
CFG = {**CONFIG, 'RUN_UI': False, 'RUN_PIPELINE': False}
```

UI

- To run UI locally (in proper environment):

```
import streamlit as st # installed in your env
```

```
# streamlit run your_notebook_app.py (if exported)
```

or run the notebook cell that calls run_ui(tracker, get_state, get_actions, cr
RUN_UI guard.

ML bridge

- Provide bundle or dataset: /kaggle/input/ed-pipeline-bundle-
ui/ed_phase2_model_thr_patched(1).joblib

- MIMIC-IV-ED demo path (for feature shim): /kaggle/input/mimic-iv-demo-v2-2/*.csv

Paths & Files (Current Freeze)

Key paths (example from last run):

- EQUIPMENT_STATUS_PATH: /kaggle/working/equipment_status.csv
- EQUIPMENT_MOVES_LOG_PATH: /kaggle/working/moves_log.csv
- SOP_REGISTRY_PATH: /kaggle/working/sop_registry.csv
- QR_OUTPUT_DIR: /kaggle/working/qrs
- EVENT_LOG_PATH: /kaggle/working/event_log.jsonl
- MODEL_BUNDLE_PATH: autodetected to /kaggle/working/ed_phase2_model_thr_patched(1).
- Advice outputs: /kaggle/working/advice_gatepos_mlp.csv (advice-only), /kaggle/working/advice_ops_panel.csv (optional merge).

Quickchecks (Optional, Non-Destructive)

Minimal function you can keep for sanity (overlay; call only when needed):

```
def phase1_quickcheck(CONFIG):

    from pathlib import Path; import inspect, pandas as pd

    CFG = dict(CONFIG); CFG['RUN_UI']=False; CFG['RUN_PIPELINE']=False

    for k in ('QR_OUTPUT_DIR', 'EVENT_LOG_PATH', 'SOP_REGISTRY_PATH', 'EQUIPMENT_STATUS_
T_MOVES_LOG_PATH'):

        p = Path(CFG[k]); (p.parent if p.suffix else p).mkdir(parents=True, exist_ok=

    try: from tracker_core import TrackerService, QRService, SOPRegistry
except Exception: pass # in-notebook classes may already be present

    t = TrackerService.from_config(CFG)

    t.log_move('__qc__', '', 'ZZ')

    sop_ok = isinstance(refresh_sop_registry(CFG, base_url='http://127.0.0.1:9/offlin

    ui_ok = True # after split, 'Vitals overdue (min)' is in run_ui_header_controls

    return {'moves': True, 'sop': sop_ok, 'ui_vitals_threshold': ui_ok}
```

Known Limits & Next Steps

Limits

- Demo uses MIMIC-IV-ED demo subset; ops toggles default to 0 in the shim.
- Sklearn version mismatch warnings during bundle load (acceptable for demo).
- UI launch depends on Streamlit in the runtime; offline PDF/SOP rendering is basic.

Next Steps

- Phase 2.5–3: multi-task MLP heads for 5 clinical targets (advice-only), calibrate & integrate with gate engine as refiners.
- Add lightweight usage telemetry for SOPs (access counts).
- Optional: persist UI thresholds to CONFIG/ENV for session restore.